

EXP2

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Course Name: Computer Networks Lab

Course Code: BCSE308L

AIM:

To capture live network traffic using Wireshark and analyze packet details.

To apply capture and display filters to identify protocols like ARP, ICMP, TCP, UDP, and TLS.

To analyze captured packets using Scapy to extract statistics such as IPs, protocols, and packet counts.

5 common protocols:

TCP

No.	Time	Source	Destination	Protocol	Length	This is an ACK to the segment in frame	The RTT to ACK the segment was	Info
4567	24.828411	2401:4900:7b78:35fc::	2603:1063:27:1::14	TCP	1329			57752 → 443 [ACK] Seq=531 Ack=8656 Win=65024 Len=1255 [TCP
4568	24.866480	2401:4900:7b78:35fc::	2603:1063:27:1::14	TCP	86			57752 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK
6186	32.460468	10.77.167.47	10.77.167.180	TCP	54	6183		0.000098000 57984 → 53 [ACK] Seq=1 Ack=1 Win=65280 Len=0
6707	32.609641	10.77.167.47	10.77.167.180	TCP	54	6664		0.042244000 57984 → 53 [ACK] Seq=31 Ack=2 Win=65280 Len=0
6720	32.633618	10.77.167.47	10.77.167.180	TCP	54	6718		0.000153000 57984 → 53 [ACK] Seq=32 Ack=57 Win=65280 Len=0
6713	32.631110	10.77.167.47	10.77.167.180	TCP	54	6710		0.000935000 57984 → 53 [FIN, ACK] Seq=31 Ack=56 Win=65280 Len=0
6191	32.462141	10.77.167.47	10.77.167.180	TCP	56			57984 → 53 [PSH, ACK] Seq=1 Ack=1 Win=65280 Len=2 [TCP PDU
6134	32.454788	10.77.167.47	10.77.167.180	TCP	66			57984 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK
731	0.951907	10.77.167.47	23.205.118.16	TCP	54	730		0.000108000 58105 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
737	0.990155	10.77.167.47	23.205.118.16	TCP	54	736		0.000025000 58105 → 80 [ACK] Seq=113 Ack=189 Win=65280 Len=0
735	0.986366	10.77.167.47	23.205.118.16	TCP	54	734		0.000167000 58105 → 80 [FIN, ACK] Seq=112 Ack=188 Win=65280 Len=0
729	0.925623	10.77.167.47	23.205.118.16	TCP	66			58105 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK
756	2.104145	2401:4900:7b78:35fc::	2600:140f:4200::17c::	TCP	74	755		0.000071000 58106 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
761	2.128769	2401:4900:7b78:35fc::	2600:140f:4200::17c::	TCP	74	760		0.000047000 58106 → 80 [ACK] Seq=113 Ack=189 Win=65280 Len=0
762	2.129901	2401:4900:7b78:35fc::	2600:140f:4200::17c::	TCP	74			58106 → 80 [FIN, ACK] Seq=113 Ack=189 Win=65280 Len=0
754	2.041883	2401:4900:7b78:35fc::	2600:140f:4200::17c::	TCP	86			58106 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK
5691	30.079084	2401:4900:7b78:35fc::	2603:1063:14:1:12a	TCP	74	5689		0.048761000 58319 → 443 [ACK] Seq=51 Ack=40 Win=254 Len=0
2140	13.318982	2401:4900:7b78:35fc::	2603:1040:5:3::a	TCP	74	2139		0.044374000 58320 → 443 [ACK] Seq=52 Ack=28 Win=252 Len=0
5673	29.408832	2401:4900:7b78:35fc::	2603:1040:5:3::a	TCP	74	5671		0.048248000 58320 → 443 [ACK] Seq=91 Ack=55 Win=252 Len=0
5736	31.550485	10.77.167.47	10.77.167.180	TCP	54	5733		0.000009000 58509 → 53 [ACK] Seq=1 Ack=1 Win=65280 Len=0
5754	31.616598	10.77.167.47	10.77.167.180	TCP	54	5751		0.045348000 58509 → 53 [ACK] Seq=32 Ack=2 Win=65280 Len=0
5358	31.638823	10.77.167.47	10.77.167.180	TCP	54	5355		0.000030000 58509 → 53 [ACK] Seq=32 Ack=2 Win=65280 Len=0

UDP

1	udpcap	254	2401:4900:7b78:35fc::	2606:4700:83b2:7cbc::	QUIC	143		0-RTT, DCID=3bbcacc8722945d7
1	udplite	003	2401:4900:7b78:35fc::	2001:4860:4826:7700::	QUIC	139		0-RTT, DCID=7b7a1fab04a3eb73
4608	25.050362	2401:4900:7b78:35fc::	2404:6800:4007:82f::	QUIC	141		0-RTT, DCID=abcc292bf3b525d4	
4610	25.050362	2401:4900:7b78:35fc::	2404:6800:4007:82f::	QUIC	1288		0-RTT, DCID=abcc292bf3b525d4	
4611	25.050378	2401:4900:7b78:35fc::	2404:6800:4007:82f::	QUIC	1078		0-RTT, DCID=abcc292bf3b525d4	
6716	32.633445	2401:4900:7b78:35fc::	2404:6800:4007:805::	QUIC	144		0-RTT, DCID=d30cbf2d7282e4b	
4739	25.237222	2401:4900:7b78:35fc::	2404:6800:4003:c24::	QUIC	142		0-RTT, DCID=dd0409ce230b4fd1	
4740	25.237608	2401:4900:7b78:35fc::	2404:6800:4003:c24::	QUIC	749		0-RTT, DCID=dd0409ce230b4fd1	
4741	25.237720	2401:4900:7b78:35fc::	2404:6800:4003:c24::	QUIC	168		0-RTT, DCID=dd0409ce230b4fd1	
743	1.285088	2001:4860:4827:7700::	2401:4900:7b78:35fc::	UDP	94		443 → 56175 Len=28	
764	5.042849	2001:4860:4827:7700::	2401:4900:7b78:35fc::	UDP	137		443 → 56175 Len=71	
745	1.304364	2404:6800:4007:83e::	2401:4900:7b78:35fc::	UDP	89		443 → 56454 Len=23	
747	1.330194	2404:6800:4007:83e::	2401:4900:7b78:35fc::	UDP	92		443 → 56454 Len=26	
740	1.192975	2404:6800:4007:83e::	2401:4900:7b78:35fc::	UDP	99		443 → 56454 Len=33	
744	1.298148	2404:6800:4007:83e::	2401:4900:7b78:35fc::	UDP	142		443 → 56454 Len=76	
3277	14.375960	2404:a800:6:147::c	2401:4900:7b78:35fc::	UDP	1118		443 → 65334 Len=1052	
3631	22.419905	2404:a800:6:147::c	2401:4900:7b78:35fc::	UDP	1165		443 → 65334 Len=1099	
575	0.398309	2404:a800:6:147::c	2401:4900:7b78:35fc::	UDP	1172		443 → 65334 Len=1106	
1035	8.803334	2404:a800:6:147::c	2401:4900:7b78:35fc::	UDP	1172		443 → 65334 Len=1106	
2392	14.240331	2404:a800:6:147::c	2401:4900:7b78:35fc::	UDP	1172		443 → 65334 Len=1106	
6	0.081228	2404:a800:6:147::c	2401:4900:7b78:35fc::	UDP	178		443 → 65334 Len=112	
5305	0.978935	2404:a800:6:147::c	2401:4900:7b78:35fc::	UDP	178		443 → 65334 Len=112	

Frame 5771: Packet, 143 bytes on wire (1144 bits), 143 bytes captured (1144 bits)

Ethernet II, Src: Intel_06:de:74 (dc:45:46:06:4e:74), Dst: 3e:7c:76:c1:1a:c2 (3e:7c:76:c1:1a:c2)

Internet Protocol Version 6, Src: 2401:4900:7b78:35fc:a902:9ac3:f25f:7347, Dst: 2606:4700:83b2:7cbc:c2a5

User Datagram Protocol, Src Port: 51989, Dst Port: 443

QUIC IE

0000 3e 7c 76 c1 1a c2 dc 45 46 06 4e 74 86 dd 60 09 >|v... E F N T ...

0010 01 05 00 59 11 3f 24 01 49 00 7b 78 35 fc a9 02 >...Y.?.? I {x5...

0020 9a c3 f2 5f 73 47 26 06 47 00 83 b2 7c bc c2 a5 >...s68 G ...

0030 06 27 5f f2 75 c4 cb 15 01 bb 00 59 d4 46 de 00 >...u...Y.f...E...00

0040 00 00 01 08 3b bc ac c8 72 29 45 d7 00 40 40 f8 >...;...r)E...00

0050 71 0d 2a ed ad 46 12 d6 49 d9 eb ff c8 27 fd 53 >q*...F...I...S

0060 e5 df 57 9b 4d f6 48 b2 c3 6c 3b 91 fb 3a fc 79 >...W.M.H...l;...y

0070 64 01 90 96 77 07 f7 03 01 cc 22 47 e2 97 23 18 >...d...w...G...#

0080 42 33 ec 7b db 53 d1 e7 b0 0b 5c 10 f5 13 57 >B3 {S...A...W

TLS

No.	Time	Source	Destination	Protocol	Length	This is an ACK to the segment in frame	The RTT to ACK the segment was	Info
5676	29.756968	44.235.12.31	10.77.167.47	TLSv1.3	78			Application Data
5688	29.892752	2401:4900:7b78:35fc::	2603:1063:14::12a	TLSv1.2	124			Application Data
5689	30.030323	2603:1063:14::12a	2401:4900:7b78:35fc::	TLSv1.2	113	5688	0.137571000	Application Data
3354	19.561682	44.235.12.31	10.77.167.47	TLSv1.3	668			Application Data, Application Data, Application Data
3364	19.759846	44.235.12.31	10.77.167.47	TLSv1.3	668			Application Data, Application Data, Application Data
3356	19.562377	10.77.167.47	44.235.12.31	TLSv1.3	118			Change Cipher Spec, Application Data
3366	19.768707	10.77.167.47	44.235.12.31	TLSv1.3	118			Change Cipher Spec, Application Data
4563	24.667273	2401:4900:7b78:35fc::	2603:1063:27:1::14	TLSv1.3	154			Client Hello (SNI=ecs.office.com)
4553	24.608745	2401:4900:7b78:35fc::	2603:1063:27:1::14	TLSv1.3	524			Client Hello (SNI=go-updater.brave.com)
3343	19.267814	10.77.167.47	44.235.12.31	TLSv1.3	790			Client Hello (SNI=go-updater.brave.com)
3348	19.496023	10.77.167.47	44.235.12.31	TLSv1.3	726			Client Hello (SNI=go-updater.brave.com)
5770	31.715226	2401:4900:7b78:35fc::	2606:4700:83b2:7cbc::	QUIC	1292			Initial, DCID=f0b0409ce230b4fd1, PKN: 3, CRYPTO, PING, PADDING
3313	15.105972	2401:4900:7b78:35fc::	2404:6800:4826:7700::	QUIC	1292			Initial, DCID=f0b0409ce230b4fd1, PKN: 3, CRYPTO, CRYPTO, PING
4608	25.050097	2401:4900:7b78:35fc::	2404:6800:4807:82f::	QUIC	1292			Initial, DCID=abcc292fb3b525da, PKN: 2, CRYPTO, CRYPTO, PING
6715	32.633133	2401:4900:7b78:35fc::	2404:6800:4807:805::	QUIC	1292			Initial, DCID=d30cbf2d7282e4b, PKN: 2, PING, CRYPTO, PADDING
4738	25.237093	2401:4900:7b78:35fc::	2404:6800:4803:c24::	QUIC	1292			Initial, DCID=dd0409ce230b4fd1, PKN: 3, PADDING, CRYPTO, PING
5776	31.030763	2606:4700:83b2:7cbc::	2401:4900:7b78:35fc::	QUIC	1266			Initial, SCID=01bca88da4b96d107ebc96886ab94b0df9474493, PKN: 5, CRYPTO, PADDING
4659	25.186368	2404:6800:4807:82f::	2401:4900:7b78:35fc::	QUIC	1296			Initial, SCID=ebcc292fb3b525da, PKN: 3, CRYPTO, PADDING
6735	32.694156	2404:6800:4807:805::	2401:4900:7b78:35fc::	QUIC	1292			Initial, SCID=f30cbf2d7282e4b, PKN: 3, CRYPTO, PADDING
3320	15.192970	2001:4860:4826:7700::	2401:4900:7b78:35fc::	QUIC	1296			Initial, SCID=fb7a1fabed4a3eb73, PKN: 5, CRYPTO, PADDING
4756	25.335901	2404:6800:4803:c24::	2401:4900:7b78:35fc::	QUIC	1296			Initial, SCID=f0b0409ce230b4fd1, PKN: 5, CRYPTO, PADDING

ICMP

No.	Time	Source	Destination	Protocol	Length	This is an ACK to the segment in frame	The RTT to ACK the segment was	Info
7526	40.039007	fe80::3c7c:76ff:fec::	2401:4900:7b78:35fc::	ICMPv6	86			Neighbor Solicitation for 2401:4900:7b78:35fc:a902:9ac3:f25f:7347
7529	40.039418	2401:4900:7b78:35fc::	fe80::3c7c:76ff:fec::	ICMPv6	86			Neighbor Advertisement 2401:4900:7b78:35fc:a902:9ac3:f25f:7347 (s

Frame 7526: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface \Device\NPF_{...}	0000	dc 45 46 06 4e 74 3e 7c	7c c1 1a c2 86 dd 60 00	EF NetS I {x5...
Ethernet II, Src: Intel_06:4e:74 (dc:45:46:06:4e:74), Dst: Intel_06:4e:74 (dc:45:46:06:4e:74)	0010	00 00 00 20 3a ff fa	00 00 00 00 00 00 3c 7c	<[
Internet Protocol Version 6, Src: fe80::3c7c:76ff:fec1:1ac2, Dst: 2401:4900:7b78:35fc:a902:9ac3:f25f:7347	0020	76 ff fe c1 1a c2 24 01	49 00 7b 78 35 fc a9 02	v.....I {x5...
Internet Control Message Protocol v6	0030	9a c3 f2 5f 73 47 87 00	4c 5c 00 00 00 00 24 01	...sG...L\{x5...
	0040	49 00 7b 78 35 fc a9 02	9a c3 f2 5f 73 47 01 01	I {x5...sG...
	0050	3e 7c 7c c1 1a c2		>[v.....sG...

ARP

No.	arp	Time	Source	Destination	Protocol	Length	This is an ACK to the segment in frame	The RTT to ACK the segment was	Info
5144	38.001397	3e:7c:76:c1:1a:c2	Intel_06:4e:74	ARP	42				Who has 10.77.167.47? Tell 10.77.167.180
5145	38.001414	Intel_06:4e:74	3e:7c:76:c1:1a:c2	ARP	42				10.77.167.47 is at dc:45:46:06:4e:74

▶ Frame 5145: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{...} ▶ Ethernet II, Src: Intel_06:4e:74 (dc:45:46:06:4e:74), Dst: 3e:7c:76:c1:1a:c2 (3e:7c:76:c1:1a:c2) ▶ Address Resolution Protocol (reply)	0000 3e 7c 76 c1 1a c2 dc 45 46 06 4e 74 08 06 00 01 0010 08 00 06 04 00 02 dc 45 46 06 4e 74 0a 4d a7 2f 0020 3e 7c 76 c1 1a c2 0a 4d a7 b4	> v...E F Nt...E F Nt M/ > v...M...
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LAYERS IN PROTOCOLS:

1. ARP:

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▶ Frame 5144: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{...}
▶ Ethernet II, Src: 3e:7c:76:c1:1a:c2 (3e:7c:76:c1:1a:c2), Dst: Intel_06:4e:74 (dc:45:46:06:4e:74)
  ▶ Destination: Intel_06:4e:74 (dc:45:46:06:4e:74)
  ▶ Source: 3e:7c:76:c1:1a:c2 (3e:7c:76:c1:1a:c2)
    Type: ARP (0x0806)
    [Stream index: 0]
  ▶ Address Resolution Protocol (request)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: request (1)
    Sender MAC address: 3e:7c:76:c1:1a:c2 (3e:7c:76:c1:1a:c2)
    Sender IP address: 10.77.167.180
    Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)
    Target IP address: 10.77.167.47

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2. UDP:

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> Frame 4666: Packet, 91 bytes on wire (728 bits), 91 bytes captured (728 bits) on interface \Device\NPF_{6600A9D0-F702-4A6A-BA40-647C98BB210E}, id 0
▼ Ethernet II, Src: Intel_06:4e:74 (dc:45:46:06:4e:74), Dst: 3e:7c:76:c1:1a:c2 (3e:7c:76:c1:1a:c2)
  > Destination: 3e:7c:76:c1:1a:c2 (3e:7c:76:c1:1a:c2)
  > Source: Intel_06:4e:74 (dc:45:46:06:4e:74)
  Type: IPv6 (0x86dd)
  [Stream index: 0]
▼ Internet Protocol Version 6, Src: 2401:4900:7b78:35fc:a902:9ac3:f25f:7347, Dst: 2001:4860:4826:7700::
  0110 .... = Version: 6
  > .... 0000 0000 .... .... = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)
  .... 1000 0101 1101 0101 = Flow Label: 0x85d65
  Payload Length: 37
  Next Header: UDP (17)
  Hop Limit: 63
  > Source Address: 2401:4900:7b78:35fc:a902:9ac3:f25f:7347
  > Destination Address: 2001:4860:4826:7700::
  [Stream index: 8]
▼ User Datagram Protocol, Src Port: 64816, Dst Port: 443
  Source Port: 64816
  Destination Port: 443
  Length: 37
  Checksum: 0xefa1 [unverified]
  [Checksum Status: Unverified]
  [Stream index: 8]
  [Stream Packet Number: 44]
  > [Timestamps]
  UDP payload (29 bytes)
  > Data (29 bytes)

```

3. TCP:

4. ICMP:

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> Frame 7529: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface \Device\NPF_{6600A9D0-F702-4A6A-BA40-647C98BB210E}, id 0
▼ Ethernet II, Src: Intel_06:4e:74 (dc:45:46:06:4e:74), Dst: 3e:7c:76:c1:1a:c2 (3e:7c:76:c1:1a:c2)
  > Destination: 3e:7c:76:c1:1a:c2 (3e:7c:76:c1:1a:c2)
  > Source: Intel_06:4e:74 (dc:45:46:06:4e:74)
  Type: IPv6 (0x86dd)
  [Stream index: 0]
▼ Internet Protocol Version 6, Src: 2401:4900:7b78:35fc:a902:9ac3:f25f:7347, Dst: fe80::3c7c:76ff:fec1:1ac2
  0110 .... = Version: 6
  > Frame 4670: Packet, 58 bytes on wire (464 bits), 58 bytes captured (464 bits) on interface \Device\NPF_{6600A9D0-F702-4A6A-BA40-647C98BB210E}, id 0
▼ Ethernet II, Src: 3e:7c:76:c1:1a:c2 (3e:7c:76:c1:1a:c2), Dst: Intel_06:4e:74 (dc:45:46:06:4e:74)
  > Destination: Intel_06:4e:74 (dc:45:46:06:4e:74)
  > Source: 3e:7c:76:c1:1a:c2 (3e:7c:76:c1:1a:c2)
  Type: 802.1Q Virtual LAN (0x8100)
  [Stream index: 0]
▼ 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 0
  000. .... = Priority: Best Effort (default) (0)
  ...0 .... = DEI: Ineligible
  .... 0000 0000 0000 = ID: 0
  Type: IPv4 (0x0800)
▼ Internet Protocol Version 4, Src: 52.66.184.244, Dst: 10.77.167.47
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0xb8 (DSCP: EF, ECN: Not-ECT)
  Total Length: 40
  Identification: 0x4902 (18690)
  > 010. .... = Flags: 0x2, Don't fragment
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 245
  Protocol: TCP (6)
  Header Checksum: 0x9d62 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 52.66.184.244
  Destination Address: 10.77.167.47
  [Stream index: 9]
  > Transmission Control Protocol, Src Port: 443, Dst Port: 55594, Seq: 6146, Ack: 52661, Len: 0

```

5.

